# STRUCTURAL NOTES & DESIGN CRITERIA

SHAMANDEEL, QUESNA, MENOUFIA, EGYPT · ECP 201 — Egyptian Code for Loads · ECP 203 — Egyptian Code for Reinforced Concrete

## WHAT THIS PACKAGE IS

The frame set out: a grid, twelve column coordinates, six slab outlines, the spans between them and the lines where a beam is required. All of it falls out of the architecture and all of it is exact.

It exists so the structural engineer starts from a coordinated layout rather than from nothing, and so the architecture cannot quietly move underneath the structure.

## WHAT THIS PACKAGE IS NOT

It is not a structural design. No member size, no slab or footing depth, no reinforcement, no cover, no load and no analysis appears anywhere in it.

Design criteria — dead and imposed loads, wind, the seismic zone and its design ground acceleration — are set and signed by the structural engineer to ECP 201, and the concrete to ECP 203. ECP 201 has five seismic zones; which governs Shamandeel is named by the engineer, not assumed here.

## WHAT IT WAITS ON

GEO-001, the soil investigation. The ground is central Nile Delta and well characterised as a region, which is enough to assume a foundation type and lay it out, and not enough to size one. Allowable bearing pressure, the water table and sulphate content are plot figures.

SUR-001, the measured survey, for the boundary STR-400 sits on.

## THE ONE RULE THE ARCHITECTURE ALREADY SETTLED

A 300 mm wall needs a beam under it, and a beam is a column line, a wall below, or a secondary beam landing on primary beams. Every 300 mm wall in this building stands on a grid line, with one exception: the penthouse's north wall, which is a line load on the +13.20 slab spanning B to C. STR-305 carries a secondary beam under it for that reason.

The 120 mm partitions do not stand on grid lines and do not need to; they are carried by the slab.

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LEVEL the whole frame · NTS · METRES

## KEY

| | |
|---|---|
| ▬ | Masonry wall cut by section — 200 mm partition, 300 mm external / party |
| ▬ | Internal stud or block partition, 120 mm |
| ▬ | Terrace parapet, 250 mm × 1100 mm high |
| ⌒ | Single leaf door, swing shown to opening side |
| ▬ | Glazed opening — arched head where noted |
| ┄ | Element above the cut plane (pergola, slab over) |
| ▭ | Void / open to sky |
| ▽ +0.00 | Finished floor level datum |

## THE FRAME

| | |
|---|---|
| GRID | 3 × 4, A–C / 1–4 |
| SPAN A–B | 4.81 m |
| SPAN B–C | 4.39 m |
| SPAN 1–2 | 4.81 m |
| SPAN 2–3 | 4.10 m |
| SPAN 3–4 | 4.29 m |
| COLUMNS | 12 |
| COLUMN SECTION | 300 × 300 mm |
| STRUCTURAL ZONE | 300 mm |
| STOREY HEIGHT | 3.30 m |

## GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. Design criteria to ECP 201 and concrete to ECP 203, set and signed by the structural engineer. No value is assumed here.

4. The package waits on GEO-001 for anything below ground and on SUR-001 for the boundary.

**FIVE-STOREY VACATION VILLA** — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

**STRUCTURAL NOTES & DESIGN CRITERIA**

| DRAWING No | REVISION | SCALE | DATE |
|---|---|---|---|
| **STR-001** | **P19** | **NTS** | **2026-10-07** |

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P16 2026-10-07 neighbours as surveyed: west 3 storeys and blind, east single storey
P17 2026-10-07 ARC-000 deliverables list; title corrected to five storeys
P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor